TEXORA · LECTURE SHEET

Fineness and Maturity of Cotton

A cotton fibre is a single plant cell: a hollow tube whose wall thickens as the boll matures. Fineness describes the tube's size, maturity describes how completely its wall was built — and the micronaire reading, used with care, speaks about both at once.

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1. The cotton fibre's structure — the basis of both properties

A cotton fibre grows from the seed coat in two stages. In the first (about 15–25 days after flowering) the cell elongates into a thin-walled tube — the **primary wall** — reaching its full length and perimeter. In the second stage, layers of cellulose (the **secondary wall**) are deposited inside the primary wall, day by day, until the boll opens. When the boll opens the fibre dries, the central channel (**lumen**) collapses, and the fibre becomes the familiar twisted ribbon with convolutions.

- **Fineness** is set mainly in stage one — by the perimeter the tube reached. It is a varietal, hereditary property: fine cottons have small perimeters (and usually long fibres); coarse cottons have large perimeters.
- **Maturity** is set in stage two — by how much cellulose was laid down relative to that perimeter. A fibre whose wall thickened fully is **mature (normal)**; one with an incomplete wall is **thin-walled (immature)**; one that stopped growing early, with almost no secondary wall, is **dead**.

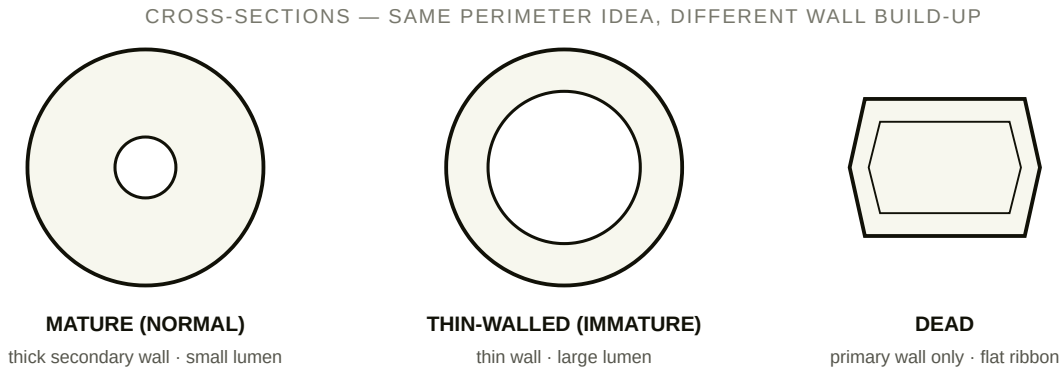


Figure 1 — Cotton maturity in cross-section. Maturity is the degree of cell-wall thickening relative to the fibre's perimeter — not the fibre's size.

2. Why the two properties matter — separately

PROPERTY	EFFECTS IN PROCESSING AND PRODUCT
Fineness (linear density, e.g. mtex)	More fibres per yarn cross-section → evener, stronger yarn and finer spinnable counts; finer fibres give softer handle and better cover, but entangle more readily into neps.
Maturity	Immature fibres are weak, break into short fragments, tangle into **neps** (Topic 6), dye paler than mature fibres (causing white-speck / barré faults in dyed fabric), and lower yarn strength. Dead fibres are the extreme case.

The commercial trap: a low-micronaire cotton may be genetically fine *and* mature (valuable) or genetically coarse *but immature* (troublesome). Micronaire alone cannot tell them apart — see §4.

3. Measuring fineness

- **Gravimetric (direct) method.** A counted number of fibres of measured total length is weighed on a microbalance; linear density = mass / length, reported in **millitex (mtex)**. Cotton typically ranges about 120–240 mtex (very fine ≈ 120, coarse ≈ 240+). Slow, but the reference definition of fineness.
- **Airflow methods** — the micronaire instrument (§4) and instruments such as the Shirley Fineness/Maturity Tester, which measures airflow through plugs at two compressions to separate fineness from maturity (the "double-compression" principle: at the second, tighter compression the effect of wall thickness changes relative to perimeter, letting the two be solved separately).
- **AFIS** (Topic 9) reports fineness in mtex from its individual-fibre measurements.

4. The micronaire test — and what it really measures

In the **micronaire** instrument a fixed mass of cotton (historically 50 grains = 3.24 g in the original instrument; modern HVI uses its own plug) is compressed into a chamber of fixed volume and air is passed through at a fixed pressure. The flow rate — read as **micronaire units** on a calibrated scale — falls as the fibres' total surface area rises. Surface area per gram depends on *both* perimeter (fineness) *and* wall thickness (maturity): a thin-walled fibre collapses into a flatter ribbon and presents more surface. Hence:

Micronaire reading = f (fineness, maturity) — a combined index, not a pure fineness

MICRONAIRE RANGE	INTERPRETATION (TRADE)
Below 3.0	Very fine — or immature; investigate before buying cheaply
3.0 – 3.9	Fine
4.0 – 4.9	Medium — the preferred commercial range (base range often 3.5–4.9)

5.0 – 5.9	Coarse
6.0 and above	Very coarse

5. Measuring maturity

5.1 Caustic soda swelling test (reference method)

Fibres are swollen in 18 % sodium hydroxide solution, which rounds the fibres out: mature fibres swell into smooth rods, thin-walled fibres stay flat and twisted, dead fibres remain flat ribbons. Under the microscope (×~150–200) at least 300–500 fibres are classified as **normal (N)**, **thin-walled (T)** or **dead (D)**, and the **maturity ratio** is calculated:

$$\text{Maturity ratio} \quad M = [(N - D) / 200] + 0.7 \quad (N, D \text{ in } \%)$$

By definition a fully mature reference cotton has M = 1.0; typical commercial cottons lie between about 0.75 and 1.0 (an equivalent expression, **maturity index**, is used by AFIS on the same idea).

WORKED EXAMPLE — MATURITY RATIO FROM A SWELLING TEST

Counts: normal 68 %, dead 12 %, thin-walled 20 % (check: 68 + 12 + 20 = 100 %).
M = [(68 − 12) / 200] + 0.7 = 56/200 + 0.7 = 0.28 + 0.7 = **0.98** — a well-matured cotton.
Contrast a drought-stressed crop: normal 40 %, dead 25 %, thin-walled 35 % → M = [(40 − 25)/200] + 0.7 = 0.075 + 0.7 = **0.78** — expect neps, weak yarn and dyeing specks, whatever its micronaire says.

5.2 Other methods

- **Polarised-light microscopy** — mature wall cellulose gives characteristic interference colours; fast classification by trained operators.
- **Differential dyeing** — a two-dye mixture (classically a red and a green direct dye) colours mature fibres predominantly one shade and immature fibres the other; useful as a bulk check on fabric faults.
- **AFIS / double-compression airflow** — instrumental maturity index for routine work, calibrated against the swelling test.

6. Worked interpretation — separating fineness from maturity

WORKED EXAMPLE — TWO COTTONS, ONE MICRONAIRE

> Cottons A and B both read micronaire 3.4. Double-compression testing gives: A — fineness 155 mtex, maturity ratio 0.95. B — fineness 195 mtex, maturity ratio 0.72.
> A is a genuinely *fine, mature* cotton: it will spin fine counts evenly. B is a *coarse but immature* cotton whose low micronaire is created by its collapsed thin-walled fibres: it will nep, break down and dye unevenly. Bought on micronaire alone at the same price, B is the expensive mistake. The rule for the buyer: **never interpret micronaire without a maturity figure.**

Glossary

Primary wall — the original thin outer wall of the cotton cell. **Secondary wall** — cellulose layers deposited inside it during maturation. **Lumen** — the central channel of the fibre. **Maturity** — degree of secondary-wall thickening relative to perimeter. **Dead fibre** — a fibre with essentially no secondary wall. **Micronaire** — airflow index combining fineness and maturity. **Maturity ratio** — M = [(N−D)/200] + 0.7 from the caustic-soda classification. **mtex** — millitex; mg per km; the unit of fibre linear density.

Key points

• Fineness is set by the fibre's perimeter (hereditary); maturity by wall thickening (growing conditions). They vary independently.
• Immature fibres cause neps, weakness and pale dye specks; fine mature fibres enable fine even yarn.
• Micronaire is an airflow index of fineness *and* maturity combined — never read it as pure fineness.
• Reference maturity method: 18 % NaOH swelling + microscopic classification → maturity ratio (good cottons ≈ 0.85–1.0).
• Cotton fineness is quoted in mtex; the gravimetric method is the reference, airflow the routine.

Practice

1. A swelling test classifies 300 fibres as: normal 195, thin-walled 75, dead 30. Convert to percentages and find the maturity ratio.
2. Explain, using the two-stage growth of the fibre, how a cotton can be coarse yet immature.
3. A lot reads micronaire 2.8 and is offered at a discount as "very fine". What two tests do you order before accepting, and what result would make you refuse it?
4. Why do immature fibres dye paler than mature fibres of the same variety?

Answer key

1. N = 195/300 = 65 %; T = 25 %; D = 10 %. M = [(65 − 10)/200] + 0.7 = 0.275 + 0.7 = **0.98** (0.975 before rounding) — well matured.
2. Perimeter — hence coarseness — is fixed in the elongation stage; wall thickening happens later. Poor conditions in the second stage (drought, early frost, boll damage) leave a large-perimeter tube with a thin wall: coarse in size, immature in structure.
3. A fineness measurement that is independent of maturity (double-compression airflow or AFIS fineness in mtex) and a maturity measurement (swelling test / AFIS maturity index). Refuse if fineness is not actually fine (e.g. ≥ 190 mtex) *and* maturity ratio is low (e.g. < 0.80): the low micronaire is then an immaturity artefact.
4. Dye is taken up in the secondary-wall cellulose. A thin-walled fibre contains far less cellulose per fibre to hold dye, so at the same bath it reaches a paler shade; in fabric these fibres show as light specks and streaks.
